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24/10/2023

JAWAHARLAL NEHRU UNIVERSITY
SCHOOL OF COMPUTER AND SYSTEMS SCIENCES
Final Exam
M.CA. (Winter Semester-2024)

Course Name: Computer Architecture
Course Code: CS 106

Maximum Marks: 60
Total Time: 3.0 hours

- Q1. (a) Why RISC Architecture is preferred over X86 in the present day smart phones?
(b) What are the factors affecting the performance of Computer Architecture
(c) Determine the normalized execution time using arithmetic mean and the geometric mean

Type	CPU A	CPU B	CPU C
Program P1	5	15	25
Program P2	250	100	15
Total Time			

Which one is preferred and why

(3+3+4)

- Q2. (a) Explain the concept of parallelism or pipelining in context to CPU and GPU.
(b) A quad core processor could speed up a computer by a factor of 4 but this rarely happens. Compute the percentage of program execution that needs to be distributed across all 4 cores to achieve an overall speedup of 3. Of 2. Of 1.5. Of 1.25.
(c) You have a computing task that can be divided into three parts:
1. Part A: 50% of the task can be fully parallelized.
2. Part B: 30% of the task can be parallelized.
3. Part C: 20% of the task is inherently serial and cannot be parallelized.

You want to calculate the overall speedup when using 16 processors for Part A and 8 processors for Part B. Part C remains serial.

(3+4+5)

Q3. (a) What is the difference between cache memory, flash memory and virtual? Explain the term locality of reference and spatial locality.

(b) Explain about the type of memory used in primary memories.

(c) Write an assembly program to explain different types of addressing modes in 8085 microprocessor.

(5+2+3)

Q4. (a) What are the basic cache optimization techniques?

(b) Calculate the average access time in both types of memory hierarchy for the following parameters:

- i) Clock cycle is 5 ns
- ii) Miss cycle is 10 clock cycle
- iii) 5 percent of instruction is not found in Instruction cache
- iv) 10 percent of instruction is not found in Data cache
- v) 25 percent of the total memory access for the data
- vi) Cache Access Time (including hit detection) is 1 clock cycle.

Which one of the memory configuration is better?

(3+6)

Q5. (a) What is Dynamic and a Non-linear pipelining technique?

(b) For the Multiplier and Addition reservation table

- (i) Determine all the forbidden latencies
- (ii) Show the state transition diagram
- (iii) Determine the optimum MAL
- (v) Determine the pipeline throughput with $\Delta = 20\text{ns}$
- (vi) What is the efficiency of the pipeline technique.

(3+7)

Q6. (a) Explain the features of vector processor Architecture?. What are the different Taxonomy of Parallel Architectures.

(b) What is the advantage of using multi core?. Explain the difference between loosely coupled and tightly coupled Microprocessor.